

JACOB D. CASTRO

Santa Maria, CA, USA (willing to relocate)

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Full stack engineer building web products since 2020, most recently in TypeScript, React, Node, and GraphQL. Founder and founding-engineer background, with a product engineer's habits: prototyping against real user feedback, carrying features from ambiguous idea to production, and keeping technical decisions tied to the business goal behind them.

SKILLS

Programming	TypeScript, JavaScript, Python, Rust, SQL
Frameworks	React, Node.js, Effect, GraphQL, Next.js, TanStack Query, Zod, Tailwind, Vitest, Playwright, Storybook
AI & Pipelines	LLM workflows, agent tooling, prompt engineering, structured outputs, eval harnesses, event-driven jobs
Infrastructure	PostgreSQL, Docker, GCP, AWS, Vercel, Railway, CI/CD, OpenTelemetry, Grafana, Sentry, Git, Jujutsu

EXPERIENCE

Senior Full Stack Engineer 01/2025 - Present
Phylax Systems

- Led the full stack TypeScript microservices platform: contract-first API design (130+ Zod-validated endpoints), cross-service compatibility testing, multi-tenant PostgreSQL with row-level security, React UI, and CI/CD.
- Architected an LLM agent service from the first commit: Python workflows, agent tools, generate-critique-refine loops, and deterministic output verification with grounding and hallucination checks, powering four production AI features.
- Owned core customer-facing features end to end, from ambiguous requirements through spec, implementation, and deployment, and reviewed all teammate PRs.
- Designed and built the platform's event-driven data backbone: 40+ background workers, a GraphQL-backed event indexing layer, and a typed async job engine with retry policies and payload redaction, powering alerting, real-time event processing, and AI workflows.

Co-Founder and CEO 05/2024 - 01/2025
Monea

- Built Monea's core product: a Rust-based CLI and engine for deploying rollup and Ethereum infrastructure.
- Ran the company as a technical founder while remaining its primary engineer: customer discovery, partnerships, and fundraising strategy.

Lead Engineer 11/2022 - 07/2024
Brink

- Designed, architected, built, and deployed five products, including an intents-based trading protocol and SDK, a swap interface, a block explorer, and an L2 chain.
- Owned the developer-facing surface: SDK releases to NPM, documentation, and integration support for external teams, plus conference keynotes on the protocol.

Founding Engineer 04/2022 - 08/2022
Coinbooks (YC-backed)

- Led development of a full-stack invoicing app with native integration to the core accounting product.
- Ran YC's 'write code and talk to users' playbook: met directly with paying customers, rapidly prototyped their requests, and folded their feedback into every release cycle.

Freelance Engineer 01/2020 - 01/2025
XIV Systems

- Occasional one-off client projects: full-stack feature work for Sudoswap (sudoswap.xyz), plus smart contracts, an NFT minting app, and the first e-commerce site for Highly Liquid (highlyliquidny.com), among other short-term contracts.

PROJECTS

- [Preconf Devnet Dashboard](#): ZuBerlin 2024 hackathon winner. Real-time Ethereum preconfirmation devnet dashboard.
- [CHIP-8 CPU Emulator](#): From-scratch CHIP-8 interpreter and emulator written in Rust with a minimal graphics frontend.
- [AI Agent Workflows](#): Custom CLI orchestrating automated implement, review, and ship loops with local agents.
- [EVM Calltrace](#): Call trace explorer with an Effect-based fetch layer: schema-validated decoding and retry schedules.
- [Personal Site](#): TanStack Start portfolio with Three.js galaxy navigation and a programmatic PDF resume generator.

EDUCATION

Self-taught developer and engineer. Production experience across venture-backed startups in place of a formal CS degree, with continuous project-based study in fullstack engineering, systems programming, data pipelines, and applied AI.